Chapter 1
Introduction to Oracle DBMS

CT208H – Oracle Database Management System

Content

1. Oracle 11g Enterprise Edition

Chapter 1. Introduction to Oracle DBMS

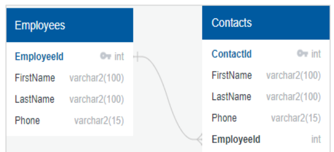
What is a database?

- A database is an organized collection of structured data stored electronically in a computer system
- Before the database system was invented, the flat file structure was commonly used to store data
  - When the number of rows in the flat file is increased e.g., million rows, it becomes unmanageable

```
first name, last name, phone
John, Doe, (408)-245-2345
Jane, Doe, (503)-234-2355
...
```

Relational database model

- Invented by Dr. Ted Codd in the 1970s
- Data is organized into entities and attributes instead of combining everything in a single structure
  - An entity (row/record) is a person, place, or thing
  - Attributes (field) describe the person, place, and thing
  - Similar entities are stored in a table
  - Related tables are stored in a database
  - Relations of entities are described by data



Relational Database Management Systems

- Database Management Systems based on relational data model (RDBMS in short)
- The most popular type of DBMS
- Some other notable ones:
  - Oracle (from Oracle Corporation)
  - DB2 (from IBM)
  - SQL Server (from Microsoft)
  - MySQL (open-source RDBMS, also from Oracle)

Oracle Database Management System

- One of the most popular RDBMS, developed by Oracle Corporation in 1977
- Oracle database characteristics:
  - Cross-platform: Windows, Unix, GNU/Linux
  - ACID-compliant: ensures data integrity and reliability
  - Oracle networking stack: applications from different platforms can communicate with each other
  - Commitment to open technologies: supports open GNU/Linux

Oracle Database Management System

- Important features:

Oracle Database Management System

- Availability:
  - Supported 24*7 availability of the database
  - Oracle Data Guard functionality: allows using of the secondary database as a copy of the primary database during any failure
- Security:
  - Unauthorized accesses are prevented
  - Oracle Advanced Security: offers Transparent Data Encryption (source level or after export) and Data Redaction (application level)

Oracle database management system

- Scalability:
  - Provides features like RAC (Real Application Cluster) and Portability, which makes an Oracle DB scalable based on usage
  - RAC provides capabilities such as rolling instance migrations, performing upgrades, maintaining application continuity, etc.
- Performance: Provides many performance optimization tools, such as Oracle Advanced Compression, and Oracle Database In-Memory,… to improve system performance

Oracle database management system

- Analytics:
  - OLAP (Oracle Analytic Processing): provides complicated analytical calculations on business data
  - Oracle Advanced Analytics: assists customers in determining predictive business models through data and text mining, as well as statistical data computation
- Management: Oracle Multitenant combines a single container database with many pluggable databases in a consolidated design

Oracle database management system

- Some other features:
  - Logical data structure: users can interact with the database without knowing where the data is stored physically
  - Partitioning: a large table can be divided into different pieces, and pieces can be stored across storage devices
  - Memory caching: allows scaling up a very large database that still can perform at a high speed
  - Portability: Oracle database can be ported on all different platforms than any of its competitors

Oracle database management system

- Some other features:
  - Backup and recovery:
    - Ensures the integrity of the data in case of system failure
    - The Recovery Manager (RMAN) tool allows DBA to perform cold, hot, and incremental database backups and point-in-time recovery
  - Clustering: enables high availability that enables the system to be up and running without interruption of services in case one or more servers in a cluster fails

Oracle database management system

- Disadvantages:
  - Complexity: Oracle is not recommended for use when the users are not technically savvy and have limited technical skills required to deal with the Oracle Database
  - Cost: the price of Oracle products is very high in comparison to other databases
  - Difficult to manage: Oracle databases are often much more complex and difficult in terms of the management of certain activities

2. Oracle Database User Types

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Types of Oracle users

- Database Administrators
- Security Officers
- Application Developers
- Application Administrators
- Database Users
- Network Administrators

Types of Oracle users

- Database administrators:
  - Oracle DBs are large and have many users
  - Main responsibilities:
    - Installing and upgrading the Oracle Server and application tools
    - Managing DBs (storage, tablespaces, DB objects)
    - Enrolling users and maintaining system security
    - Monitoring and optimizing the performance of the database
    - Planning for backup and recovery of database information

Types of Oracle users

- Application developers:
  - Design and implement database applications
  - Main responsibilities:
    - Designing the database structure for an application
    - Tuning the application during development
    - Establishing an application's security measures during development
- Application administrators: administrate a particular application

Types of Oracle users

- Security officers: concerned with enrolling users, controlling and monitoring user access to the database, and maintaining system security
- Database users: interact with the database via applications or utilities
  - entering, modifying, and deleting data, where permitted
  - generating reports of data
- Network administrators: administering Oracle networking products, such as Net8

3. SQL, PL/SQL, SQL*Plus and SQL Developer

Chapter 1. Introduction to Oracle DBMS

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SQL

- A set-based declarative language that provides an interface to an RDBMS such as Oracle Database
- A nonprocedural that describes what should be done, not how to derive it
- SQL is the ANSI standard language for relational databases.
- Oracle SQL is an implementation of the ANSI standard and supports numerous features that extend beyond standard SQL

ORACLE DB Management

PL/SQL

- A procedural extension to Oracle SQL
- Enables to use all of the Oracle Database SQL statements, functions, and data types
- Used to control the flow of SQL programs, create and use variables, and write error-handling procedures
- A primary benefit of PL/SQL is the ability to store application logic in the database itself
  => Built-in functionalities can be deployed anywhere

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SQL*Plus

- A command-line interactive user interface and batch query tool that is installed with every Oracle Database installation
- Has its own commands and environment, and it provides access to the Oracle Database
  - Format, perform calculations, store, and print from query results
  - Examine table and object definitions
  - Develop and run batch scripts
  - Perform database administration

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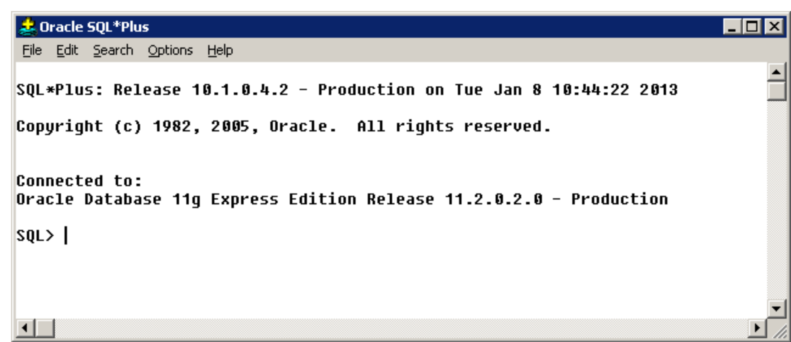
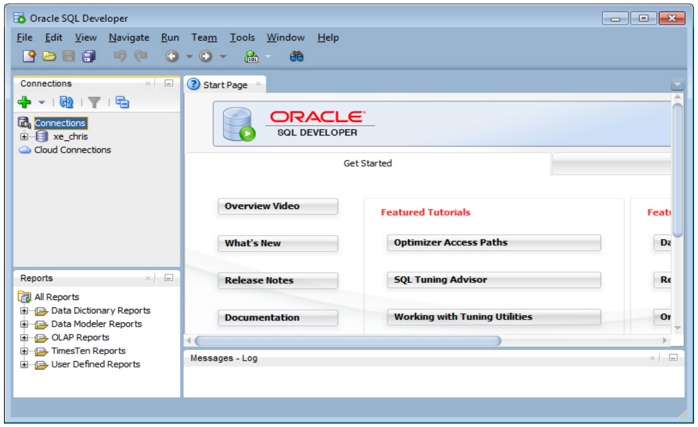
SQL*Plus



Oracle SQL Developer



4. Oracle Versions and Editions

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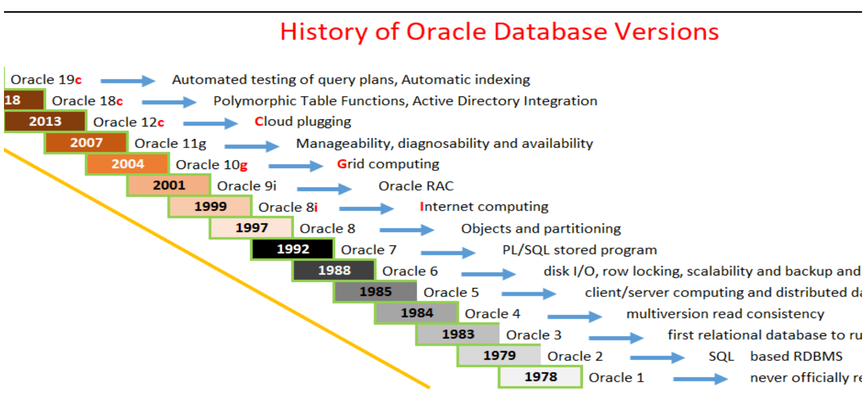
Oracle Database versions



Oracle Database 11g editions

- Enterprise Edition:
  - Provides the performance, availability, scalability, and security required for mission-critical applications such as high-volume Online Transaction Processing (OLTP) applications, query-intensive data warehouses, and demanding Internet applications
  - Contains all of the components of the Oracle Database, and can be further enhanced with the purchase of the options and packs described

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Oracle database 11g editions

- Standard Edition One:
  - Delivers unprecedented ease of use, power, and performance for workgroup, department-level, and Web applications
  - From single-server environments for small business to highly distributed branch environments
- Standard Edition:
  - Unprecedented ease of use, power, and performance of Standard Edition One
  - Supports larger machines and clustering of services with Oracle Real Application Clusters (Oracle RAC)

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Oracle database 11g editions

- Personal Edition:
  - Supports single-user development and deployment environments that require full compatibility with all other editions
  - Includes all of the components and options that are included with Enterprise Edition, with the exception of the Oracle Real Application Clusters option and the Management Packs
  - Available on Windows and Linux platforms only

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Oracle database 11g editions

- Express Edition:
  - An entry-level edition of Oracle Database
  - Quick to download, simple to install and manage, and is free to develop, deploy, and distribute
  - Can be installed on any size machine with any number of CPUs, stores up to 11GB of user data, uses up to 1 GB of memory and only one CPU on the host machine

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DB Management

5. Administration Accounts & Security

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DB administrator security and privileges

- Administrative tasks in Oracle need extra privileges both within the database and possibly in the operating system (e.g. , for software installation, executing OS commands, etc.)
- Access to a database administrator account should be tightly controlled
- Two user accounts are automatically created with the database and granted the DBA role: SYS and SYSTEM

The SYS account

- Automatically created and granted the DBA role
- All of the base tables and views for the database's data dictionary are stored in the schema SYS
- Tables in the SYS schema are manipulated only by Oracle
- Most database users should never be able to connect using the SYS account
- DBA role: contains all DB system privileges, and thus, it should only be granted to fully functional DB administrators

The SYSTEM account

- Automatically created and granted all system privileges
- The SYSTEM username creates additional tables and views that display administrative information and internal tables and views used by Oracle tools
- Never create tables of interest to individual users in the SYSTEM schema

DB administrator authentication

- DB administrators must often perform administrative operations such as shutting down or starting up a database

  => need a more secure authentication scheme than normal users
- Authentication methods:
  - Operating system authentication
  - Network authentication
  - Oracle database authentication
- Oracle allows using all methods within the same DB instance

Operating system authentication

- Benefits:
  - Once authenticated by the operating system, users can connect to Oracle more conveniently without specifying a username or password
  - With control over user authentication centralized in the operating system, Oracle need not store or manage user passwords, though it still maintains usernames in the database
  - Audit trails in the database and operating system use the same usernames

Network authentication

- Authentication capabilities at the network layer are handled by the SSL protocol or by third-party services such as:
  - DCE authentication
  - Kerberos authentication
  - Public Key Infrastructure-Based authentication
  - Authentication with RADIUS
  - Directory-based services

Oracle database authentication

- Oracle can authenticate users attempting to connect to a database by using information stored in that database
- To set up Oracle to use database authentication:
  - Creating each user with an associated password that must be supplied when the user attempts to establish a connection
  - Users' passwords are stored in the data dictionary in an encrypted format to prevent unauthorized alteration

QUESTIONS?

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